SOKOINE UNIVERSITY OF AGRICULTURE
School of Engineering and Technology

Course: ABE 721 - Agricultural Systems Modelling and Simulation
Instructor: Kadeghe Fue, PhD, P.Eng(T)

ASSIGNMENT 6
Module 6: Advanced Applications (Programming Focus)

Instructions:

- This assignment is an exclusive **computational practicum** testing your ability to translate the advanced theoretical applications of Module 6 directly into functional software code.
- **Strict Programming Requirement:** You must submit your complete, well-commented source code (Python, MATLAB, or R) along with the exact numerical results (console transcripts and plots) for *every* question below. No manual hand-calculations will be accepted.

Tier 1: Core Mathematical Algorithms

Task 1.1: Crop Phenology & Thermal Time Integrator

Significance of Task: *Translating chronological time into biological thermal time allows crop models to adapt to climate shifts and accurately predict specific harvest dates.*

Prompt: Write a script to calculate the cumulative Thermal Time ($\sum \Delta TT$) over a 10-day period.

- Use a biological base temperature $T_{base} = 10$°C.
- **Data:** Define two arrays for daily temperatures (°C): $T_{max} = [28, 29, 31, 33, 30, 27, 26, 32, 34, 30]$ and $T_{min} = [16, 17, 18, 20, 19, 15, 14, 18, 21, 18]$.
- **Logic:** Iterate through the arrays to calculate the daily ΔTT using the basic formulation: $\Delta TT = \max(0, \frac{T_{max}+T_{min}}{2} - T_{base})$.
- **Output:** Print the cumulative thermal time after 10 days. Also, determine and print exactly which day (index) the crop reaches a specific developmental stage defined by $P_1 = 120$ GDD.

Task 1.2: GIS Raster Map Algebra (USLE)

Significance of Task: *Automating raster matrix multiplication allows engineers to pinpoint extreme soil erosion hotspots instantly across massive, multi-million-pixel watersheds.*

Prompt: You are given coarse 3×3 grid representations of an agricultural watershed for the Universal Soil Loss Equation (USLE). Write a script that defines the following matrices and performs element-wise map algebra to compute the final Soil Loss matrix $\mathbf{A} = \mathbf{R} \cdot \mathbf{K} \cdot \mathbf{LS} \cdot \mathbf{C} \cdot \mathbf{P}$.

- $\mathbf{R} = [[120, 120, 120], [130, 130, 130], [140, 140, 140]]$
- $\mathbf{K} = [[0.2, 0.2, 0.4], [0.2, 0.4, 0.4], [0.4, 0.5, 0.5]]$
- $\mathbf{LS} = [[1.1, 1.5, 2.0], [1.0, 1.8, 3.5], [0.8, 2.2, 4.1]]$
- Assume $\mathbf{C} = 0.5$ and $\mathbf{P} = 1.0$ globally (as scalar multipliers).
- **Output:** Print the final 3×3 matrix $\mathbf{A}$. Programmatically locate and print the array indices (row, col) of the pixel with the absolute highest erosion risk.

Tier 2: Engineering Optimization & Kinematics

Task 2.1: CNC Kinematics via Homogeneous Transformations

Significance of Task: *Translating CAM toolpath coordinates using homogeneous 4×4 matrices is mandatory to prevent physical crashes when driving 5-axis CNC machines to manufacture complex agricultural parts.*

Prompt: Write a script to compute the final spatial position of a CNC tool tip.

- **Initial Position:** The tool is at coordinate $\mathbf{P}_0 = \begin{bmatrix} 50 \\ 20 \\ 10 \\ 1 \end{bmatrix}$ (in mm).
- **Operation 1:** A linear translation of $T_x = 10, T_y = -5, T_z = 30$. Define the 4×4 translation matrix $\mathbf{T}_{trans}$ and compute $\mathbf{P}_1 = \mathbf{T}_{trans} \cdot \mathbf{P}_0$.
- **Operation 2:** A pure rotation of exactly $\theta = 45°$ around the Z-axis. Define the 4×4 rotation matrix $\mathbf{R}_z(45°)$ and compute the final position $\mathbf{P}_2 = \mathbf{R}_z \cdot \mathbf{P}_1$.
- **Output:** Print the intermediate vector $\mathbf{P}_1$ and the final coordinate vector $\mathbf{P}_2$.

Task 2.2: Bio-Economic LP Optimization (Simplex)

Significance of Task: *Algorithmic resource optimization guarantees maximum financial profit while strictly adhering to physical land, water, and labor constraints, acting as the ultimate Decision Support System.*

Prompt: Use a Linear Programming solver (such as `scipy.optimize.linprog` in Python or `lpSolve` in R) to find the optimal crop allocation.

- **Objective:** Maximize total profit $Z = 500x_1 + 400x_2$
  (where x_1 = Hectares of Maize, x_2 = Hectares of Beans).
- **Constraints:**
  - Total Land: $x_1 + x_2 \leq 100$ hectares.
  - Irrigation Water: $400x_1 + 200x_2 \leq 32,000\ m^3$.
  - Labor Hours: $20x_1 + 30x_2 \leq 2,400$ hours.
  - Non-negativity: $x_1 \geq 0, x_2 \geq 0$.
- **Output:** Print the optimal number of hectares to plant for Maize (x_1) and Beans (x_2), and the resulting maximum total profit in TZS.

Tier 3: Soft Computing & Spatial Architecture

Task 3.1: Artificial Neural Network Feedforward Pass

Significance of Task: *Hard-coding the foundational linear algebra and non-linear activation mathematics of an ANN is the first step in utilizing AI to predict highly chaotic agricultural yields where mechanistic physics fail.*

Prompt: Code a single feedforward pass (from input to prediction) for a simple Multi-Layer Perceptron (MLP) containing an Input Layer (3 nodes), one Hidden Layer (4 nodes), and an Output Layer (1 node).

- **Inputs:** Define an environmental input vector $\mathbf{x} = [0.8, 0.4, 0.9]^T$.
- **Hidden Layer:** Generate a random 4×3 weight matrix $\mathbf{W}_1$ and a 4×1 bias vector $\mathbf{b}_1$. Calculate $\mathbf{z}_1 = \mathbf{W}_1\mathbf{x} + \mathbf{b}_1$. Apply the Sigmoid activation function $a_1 = \frac{1}{1+e^{-z_1}}$ to get the hidden node activations.
- **Output Layer:** Generate a random 1×4 weight matrix $\mathbf{W}_2$ and a scalar bias b_2. Calculate the final scalar prediction $\hat{y} = \mathbf{W}_2\mathbf{a}_1 + b_2$.
- **Output:** Print your randomly generated weights, the intermediate hidden layer activations vector $\mathbf{a}_1$, and the final scalar network prediction $\hat{y}$.

Task 3.2: Artificial Neural Network Backpropagation

Significance of Task: *Calculating error gradients and updating matrices via backpropagation is the core "learning" mechanism that allows an AI model to adapt to chaotic agricultural datasets over time.*

Prompt: Building directly upon your feedforward script from Task 3.1, write code to perform a single iteration of Gradient Descent to update the output layer weight matrix $\mathbf{W}_2$.

- **Error Calculation:** Assume the true measured yield for this data point is $y_{true} = 1.0$. Calculate the output error term: $\delta_2 = (\hat{y} - y_{true})$.
- **Gradient Calculation:** The gradient of the Cost Function with respect to the output weights is $\frac{\partial J}{\partial \mathbf{W}_2} = \delta_2 \cdot \mathbf{a}_1^T$. Compute this gradient vector.
- **Weight Update:** Apply the gradient descent update rule: $\mathbf{W}_2^{(new)} = \mathbf{W}_2^{(old)} - \alpha \frac{\partial J}{\partial \mathbf{W}_2}$. Use a learning rate of $\alpha = 0.1$.
- **Output:** Print the calculated error δ_2, the computed gradient vector, and the new updated weight matrix $\mathbf{W}_2^{(new)}$.

Task 3.3: Spatial Interpolation via Ordinary Kriging

Significance of Task: *Solving the Kriging matrix system enables the creation of continuous, high-resolution precision agriculture prescription maps from sparse, expensive field sensor data.*

Prompt: You have collected soil nitrate readings from 3 sensor locations: $Z = [45, 50, 42]^T$ ppm. You must estimate the nitrate level $\hat{Z}$ at an unsampled location. The spatial covariances between the known points and the unknown point have been pre-modeled using a theoretical semivariogram.

- **The Matrix System:** Ordinary Kriging requires solving the linear system $\mathbf{A}\lambda = \mathbf{b}$, where $\mathbf{A}$ is the semivariance matrix between known points (with Lagrange multiplier constraints) and $\mathbf{b}$ is the semivariance vector between the known points and the unknown target.
- Define $\mathbf{A} = \begin{bmatrix} 0 & 12 & 18 & 1 \\ 12 & 0 & 14 & 1 \\ 18 & 14 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 10 \\ 15 \\ 8 \\ 1 \end{bmatrix}$.
- **Logic:** Write code to mathematically invert matrix $\mathbf{A}$ (e.g., using `numpy.linalg.inv` or `solve`), and compute the optimal weight vector $\lambda = \mathbf{A}^{-1}\mathbf{b}$. The first three elements of λ are the sample weights w_1, w_2, w_3.
- **Prediction:** Calculate the estimated concentration $\hat{Z} = w_1 Z_1 + w_2 Z_2 + w_3 Z_3$.
- **Output:** Print the inverted matrix $\mathbf{A}^{-1}$, the solved weight vector λ, and the final predicted soil nitrate concentration $\hat{Z}$.

Please submit your programming scripts and their respective outputs to my email kadeghe.fue@sua.ac.tz by one week after completion of Module 6. Do not send this work to kadefue@sua.ac.tz.